

A Supervised Learning Framework for Real-Time Detection of UDP Flood and ARP Spoofing Attacks in IoT Networks

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Abstract - The growing development of contemporary networks and IoT infrastructure has led to an increase in the number of cyberattacks, such as UDP flooding and ARP spoofing attacks, which can cause a substantial decrease in network availability and communication integrity. The use of traditional signature-based Intrusion Detection Systems (IDS) is often ineffective in identifying such cyberattacks due to their static nature and inability to adapt to the dynamic nature of cyber threats. In order to overcome these issues, this research work proposes an AI-based real-time network threat analysis system that treats intrusion detection as a supervised multi-class classification task. The Edge-IIoTSet dataset, which provides a realistic representation of network flow characteristics in IoT and IIoT settings, is employed to design and evaluate the performance of a variety of supervised machine learning classifiers, including Logistic Regression, Support Vector Machine, Random Forest, Gradient Boosted Decision Trees (XGBoost), and a shallow Multi-Layer Perceptron, which this work demonstrates high detection accuracy in-domain, as per the experimental findings, with tree-based classifiers achieving a more balanced trade-off between accuracy and minority class detection. The Knowledge Discovery in Databases i.e. KDD-derived dataset serves as a sanity benchmark, and the BETH honeypot dataset serves as a cross-domain benchmark to further validate the proposed models. The existence of a large domain shift between network flow data and host-level honeypot data is shown through evaluation on the BETH dataset, while near-perfect performance on the KDD-derived dataset indicates the existence of bias in the dataset. These results imply that robustness in the real world cannot be guaranteed by high detection accuracy.

Keywords: Intrusion Detection System, UDP Flood, ARP Spoofing, Machine Learning, XGBoost, Network

Security, Domain Shift, Random Forest, Multi Layer Perceptron, Deep Learning

I. INTRODUCTION

The attack surface of modern networks has been significantly expanded by the rapid growth of the IoT and cyber-physical world, making them vulnerable to complex denial-of-service and man-in-the-middle attacks such as UDP flooding and ARP spoofing [1]. A critical safety mechanism for monitoring network traffic and rapidly detecting malicious activity is an intrusion detection system (IDS) [2]. Traditional IDS techniques, which are largely based on static signatures, tend to generate a high number of false positives and are incapable of detecting zero-day attacks [3]. Machine learning (ML)-based intrusion detection systems are now a promising substitute because of their capability to detect complex patterns in data and dynamically respond to new threats [4]. Supervised learning algorithms such as Random Forest, Support Vector Machines, and Gradient Boosting have shown outstanding detection performance on standard datasets [5], [6]. However, recent research shows that near-perfect performance on a given dataset sometimes fails to generalize to other settings, resulting in a substantial drop in performance on cross-dataset tasks [7].

Moreover, the current state of most IDS research, as used in practice, is based on artificial or outdated data sets, which makes them less applicable to modern cloud and IoT environments [8]. The heterogeneous host-level dynamics are modeled by real-world data sets such as BETH, which create challenging conditions for ML-based detection systems [9]. In an attempt to address these limitations, this research will assess the robustness and false positives of supervised IDS models on a primary and secondary IoT data set. Intelligent intrusion detection systems capable of detecting unusual behavior in real-time and comprehending complex patterns from network traffic have been enabled by recent advances in Machine Learning and Deep Learning.

Nevertheless, much of the existing research work focuses on achieving high accuracy on a single dataset, often at the expense of issues such as robustness to real-world conditions, generalization across domains, and bias in the dataset. Based on supervised learning models, this work proposes an AI-powered real-time network threat analysis approach for the detection of ARP Spoofing and UDP Flood attacks. This work bridges significant gaps in the existing literature by emphasizing robustness analysis and secondary dataset validation, aside from achieving good in-domain detection accuracy.

The main contributions of this work are given as follows:

1. Development and comparison of various supervised machine learning models for the detection of ARP spoofing and UDP flooding attacks.
2. Critical information regarding the limitations of employing network-centric IDS models.
3. Visualization and performance validation using metrics including Accuracy, Precision, Recall, F1-score, and ROC curves.

II. LITERATURE REVIEW

As a result of the increase in distributed large scale systems and the Internet Of Things (IoT), the critical function that IDS provides as an essential component to ensuring security for networks. It has received a significant amount of attention. Rahman et al. in [1] have done an analysis of IDS techniques for IoT Networks discussing increasing susceptibility to both spoofing and denial-of-service type attacks in IoT Networks [1]. Traditional rule based IDS lack scalability and adaptability within dynamic environments therefore M. Rahman A.Rahman and S.Hossain identify the need for machine learning based solutions. In [2] the detection algorithms are classified into three groups: signature-based, anomaly-based, and hybrid i.e. composed of both signature-based and anomaly-based detection algorithms. This study in [2] has indicated that signature-based systems have limitations when it comes to detecting new types of threats; furthermore, supervised machine learning methods show promise for improving the effectiveness of detecting new types of attacks within networked environments.

IDS related problems have been studied extensively by researchers in [3]. Alkassbeh, for example, evaluated a wide variety of machine learning algorithms for intrusion detection problems, including Logistic Regression, Support Vector Machine and Decision Trees and found them to be successful on benchmark intrusion detection problems. However, the properties of the dataset and the presence of redundant features had a significant impact. The ability of neural network-based IDS approaches to model complex, non-linear relationships has been explored in [4]. Di Mauro et al. have discussed neural approaches for intrusion detection and highlight the strengths and weaknesses of deep and shallow neural networks. The problems of overfitting and dependence on the dataset has been highlighted and explored, though the models perform well on small benchmark problems.

Tripathy and Behera have analysed the effectiveness of Multiple Classifier Systems i.e. MCSs as they compared the performance of several supervised machine learning algorithms in relation to IDSs. The ability of the detection system in terms of accuracy using multiple classification algorithms has been indicated and demonstrated. The effectiveness of applying Ensemble Algorithm-based classification techniques in IDSs has been confirmed; however, Tripathy and Behera in [5] also indicate that the ability of the technique to apply to real-time situations cannot be determined purely based on the accuracy of the classifier. Similarly, in [6] Onyebueke et al. demonstrated the effectiveness of using the Random Forest Classifier and the XGBoost Classifier for intrusion detection; making it clear that using Ensemble Algorithm-based classification methods will outperform Linear Classification methods when there are complex traffic patterns present.

The generalization capability of machine learning-based intrusion detection systems has been brought into question despite positive findings. Machine learning models trained on a single intrusion detection dataset tend to perform poorly on other datasets, as explicitly stated by the findings of Cantone et al. on the cross-dataset generalization capability of machine learning models in [7]. A machine learning approach for detecting DDoS attacks in the IoT context, as it relates to IoT-specific attacks, is proposed by Seifousadati et al. in [8]. Despite the challenges of real-world applicability, the work proves that supervised machine learning models are capable of accurately detecting volumetric attacks such as UDP floods in a controlled setting. Highnam et al. in [9] introduce the BETH dataset, a large honeypot-captured dataset containing real-world security events at the host level, to address the lack of realistic evaluation data. The BETH dataset is vastly different from typical network flow datasets, making it suitable for evaluating the robustness of IDS models.

III. PROPOSED SUPERVISED LEARNING FRAMEWORK FOR SPOOFING ATTACKS

Three datasets with varying features were employed to ensure a comprehensive and precise evaluation. The Edge-IIoTSet contains labeled records of entire flows and consists of labels which indicate the type of traffic i.e. cyber-attack scenarios including UDP Flood, ARP Spoofing [MITM], and benign traffic and events that occurred on an IoT or IIoT network. The Edge-IIoTSet has been chosen as the primary dataset since it represents traffic from an IoT and IIoT network. The Edge-IIoTSet provides a sufficient number of flows with a variety of flow-level features including packet counts, protocol information, timing information, transport layer information to be suitable for use in a supervised intrusion detection system. Fig.1 shows the proposed framework for the models.

Rishi Kumar Cybersecurity(Support Dataset): The Rishi Kumar cybersecurity dataset is derived from the KDD and contains labeled records for both attack and non-attack i.e. normal traffic. The high level of feature

separability and redundancy in this dataset makes it unsuitable for testing the robustness of the Jive algorithm; therefore, it was used only as a sanity check for the Jive baseline model.

BETH Dataset (Secondary Robustness Dataset): The BETH dataset contains actual honeypot data from a variety of honeypots, which consist of logs from various honeypot systems logging the bad guys at the Host log level as well as the Domain Name System level. The BETH dataset exhibits more variability from host systems in the real world than the Edge-IIoTSet dataset, which only displays network flow statistics from hosts. The Evaluation portion of the BETH dataset was used for validating cross-domain robustness and also assessing false positives with the BETH model to determine if it is possible to use retrofitting with the BETH model against other training sets to learn without retraining the BETH model.

Proposed Supervised Learning Framework for Spoofing & Flooding Detection

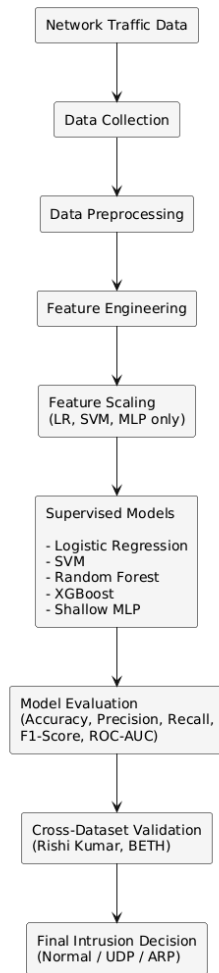


Fig.1 The proposed Framework

3.1 Dataset Preprocessing and Splitting

In order to maintain the original class distribution across regular traffic, UDP Flood, and ARP Spoofing assaults, the Edge-IIoTSet dataset was divided into training and testing subsets using an 80:20 stratified split. The sample

distribution across classes for both the entire dataset and the resulting subgroups is summarized in Table 1.

Split	Full Dataset	Training set (80%)	Testing set (20%)
Total Samples	302,409	241,927	60,482
Normal	135,620	108,496	27,124
UDP Flood	165,777	132,662	33,112
ARP Spoofing	1,012	769	246

Table 1. Data Split for Edge-IIoTSet

Dataset	Edge-IIoT Set	Rishi Kumar	BETH
Role	Primary	Support	Secondary
Classes	Normal, UDP Flood, ARP Spoofing	Normal, Attack	Benign / Malicious
Nature of Data	Network-flow (IoT/IIoT)	KDD-derived network records	Honeypot & host-level logs

Table 2. Details of Various Datasets Tested

By collecting data from various sources and reflecting different data domains, the datasets used in this investigation introduced natural distributional discrepancies. The Edge-IIoTSet dataset was used as the main training domain to build the foundation feature space. Non-informative data such as timestamps, IP addresses, host names, and label IDs were removed during preprocessing to avoid leakage and ensure model generalization. Only network-relevant features were retained for cross-dataset testing on the BETH dataset, and any missing attributes were aligned with the Edge-IIoTSet feature space using zero-padding. No feature-level retraining, adaptation, or fine-tuning was performed during cross-dataset testing in order not to disrupt the integrity of the robustness experiment.

3.2 Training and Evaluation Strategy

A multi-class classification problem was formulated as a supervised learning problem to develop the intrusion detection problem. Five classifiers of varying complexity were employed:

1. Regression Logistic
2. SVM, or support vector machine

3. The Random Forest
4. Decision Trees with Gradient Boosting (XGBoost)
5. Multi-Layer Shallow Perceptron (MLP)

Regularization techniques, early stopping, and class balancing techniques were employed as needed to counter overfitting. There were three steps to the evaluation strategy. Firstly, all models were trained, and each one was tested on a held-out test set, with Edge-IIoTSet used for both in-domain training and testing. Secondly, there was a sanity test of model behavior on a KDD-derived benchmark, which addressed the question of whether there was bias in the dataset. This was done by performing a validation test on the Rishi Kumar dataset. Finally, the top model performed on the BETH dataset without any kind of model fine-tuning to assess its robustness on a different domain. The metrics used for model evaluation on these datasets included ROC AUC, F1, recall, accuracy, precision, and anomaly trigger rate. Intrusion Detection System is based on structured network flow features to function as a supervised multi-class classification model. There are steps for feature extraction, feature representation, and classification in this modular approach. The proposed model is lightweight and suitable for real-time implementation since it only analyzes network traffic features, unlike multimedia-based systems.

Network Traffic Feature Extraction: The goal of this step is to extract valuable information from raw network traffic log data. Flow-level records obtained from network devices and edge nodes comprise the input data. Packet information, protocol-related fields, time information, and transport-layer information are some examples of fields. To prevent bias and data leakage, non-informative fields such as IP addresses, timestamps, and host information are removed. Traffic patterns relevant to volumetric and spoofing attacks, such as UDP Flood and ARP Spoofing attacks, are represented by the feature set.

Preprocessing and Feature Representation: A numerical representation suitable for supervised learning is derived from the extracted features. For models that are sensitive to the magnitude of the features, feature scaling, categorical variable encoding, and missing value imputation are applied. A common feature space is maintained across datasets to enable cross-dataset comparison. Zero-padding is applied to synchronize missing features during secondary dataset validation to ensure dimensional compatibility without retraining the models.

Models of Supervised Classification: The supervised learning models that have been trained to classify network traffic data into three categories, namely normal traffic, UDP flood attack, and ARP spoofing assault, are provided with the processed feature vectors. There are five models of supervised classification that have been implemented, each with a different complexity level. Each of these models learns different decision boundaries from the same feature space, allowing for a comparative analysis of linear, ensemble, and neural models.

Decision Fusion: An expected class label based on the individual models from the trained corpus of input examples was provided for each model individually in order to test the individual strengths and weaknesses of the two individual models. The output generated by the classifiers could be combined with auto-response or warning type methods in real life. In addition, the model architecture provides flexibility to add additional models of classification to the existing mechanism for feature extraction without changing the mechanism for feature extraction.

Architecture: The architectural design deliberately chooses to avoid complex multimodal inputs to maintain efficiency and interpretability. The method allows for the systematic study of generalization behavior on multiple datasets and succeeds in obtaining good in-domain detection performance by focusing on network flow features and supervised classification. The trade-offs between robustness, detection performance, and complexity can be assessed using multiple classifiers.

IV. EXPERIMENTAL RESULTS

Table 3 summarizes the classification performance of all five supervised models on the Edge-IIoTSet dataset. Metrics include accuracy, precision, recall, F1-score, and ROC-AUC. The macro F1-score shows variations in minority-class detection, especially for ARP spoofing, even though all models attained high accuracy. While ensemble and neural models show better balance, logistic regression shows lower recall for the minority class.

Model	Logistic Regression	SVM	Random Forest	XG Boost	MLP
Accuracy	1.00	0.98	1.00	1.00	1.00
Precision	1.00	0.72	0.93	0.91	0.97
Recall	0.77	0.99	1.00	1.00	1.00
Specificity	1.00	0.97	0.99	0.99	1.00
F1- Score	0.82	0.74	0.96	0.95	0.98
ROC - AUC	1.00	0.99	0.99	0.99	1.00

Table 3. Performance Comparison on Edge-IIoTSet

Rishi Kumar Dataset as a Sanity Benchmark: The Rishi Kumar cyber security dataset was employed as a secondary benchmark to test the separability of the datasets rather than generalization. All the models tested performed close to perfectly, even when symbolic protocol-level features are removed. This shows that the class separability is high, as expected in previous criticisms of KDD-derived intrusion detection datasets. Thus, the performance on this dataset is provided as a sanity check and not as a measure of robustness.

Cross-Dataset Assessment: BETH Dataset: When tested on the BETH honeypot dataset, the XGBoost model identified a large number of instances as anomalous. This is because there is a large domain shift from network flow-based datasets Edge-IIoTSet to host-based event logs BETH. This experiment shows the difficulty of using network-centric IDS models on diverse security telemetry without domain adaptation.

Qualitative and Visual Analysis: The class-wise classification accuracy of the Random Forest, XGBoost, and Shallow MLP models is represented through the confusion matrices in Fig.2, Fig.3, and Fig.4. Since most of the predictions are concentrated along the diagonal, it is clear that the Random Forest model has very strong discriminatory power, as evident from Fig. 2. Though ARP spoofing attacks are correctly identified with perfect recall, normal traffic and UDP flood attacks are identified with high accuracy, thereby establishing the effectiveness of the model in identifying the minority class with very few false negatives.

Likewise, the confusion matrix for the XGBoost classifier is shown in Fig. 3. With a few instances of misclassification between normal traffic and UDP Flood attacks, the classifier achieves nearly perfect results for all three classes. Notably, the ARP Spoofing instances are properly classified with no false negatives, indicating that XGBoost can potentially model interaction terms and non-linear relationships between network traffic attribute variables. High classification accuracy is also evident from the confusion matrix of the Shallow MLP model, which is shown in Fig. 4. With very slight instances of misclassification between UDP Flood and ARP Spoofing attacks, the model is able to properly classify both normal traffic and ARP Spoofing attacks. This behavior indicates that although neural networks can potentially model complex patterns effectively, slight misclassifications can still occur due to a slight overlap in attack characteristics.

Feature importance analysis provides further insights into the decision-making process of the ensemble models. Features such as `udp.stream`, `tcp.payload`, and `http.tls_port`, which are related to traffic volume and behavior, are assigned the highest importance by the Random Forest model, as shown in Fig.5. These characteristics correspond to volumetric attacks, particularly UDP Flooding, where the traffic rates are anomalous. Similarly, the most important feature in the XGBoost feature importance chart in Fig. 6 is `udp.stream`, followed by transport-layer and protocol-specific features such as `tcp.connection.synack`, `tcp.payload`, and `arp.opcode`. The ability of the model to detect protocol-level patterns related to ARP spoofing attacks is confirmed by the presence of ARP-related features in the top contributors.

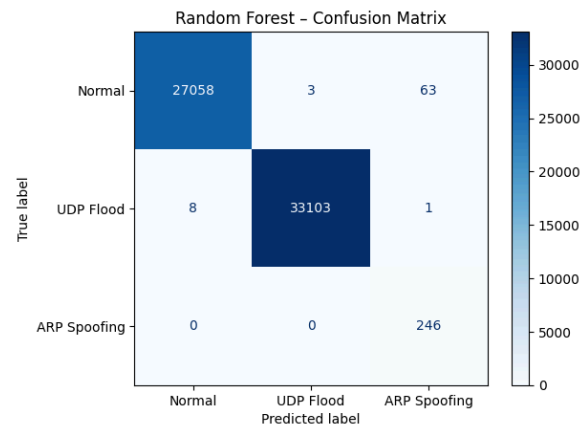


Fig.2. Confusion Matrix of Random Forest

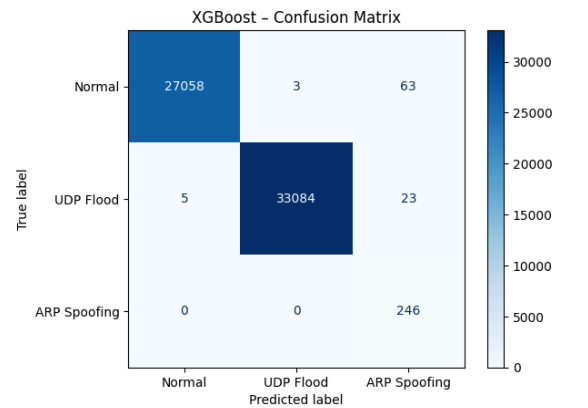


Fig.3. Confusion Matrix of XGBoost

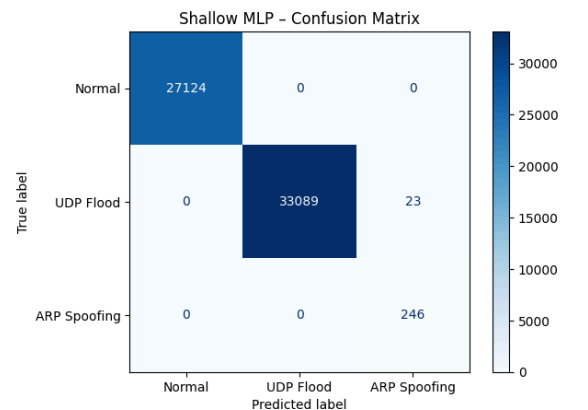


Fig.4. Confusion Matrix of Shallow MLP

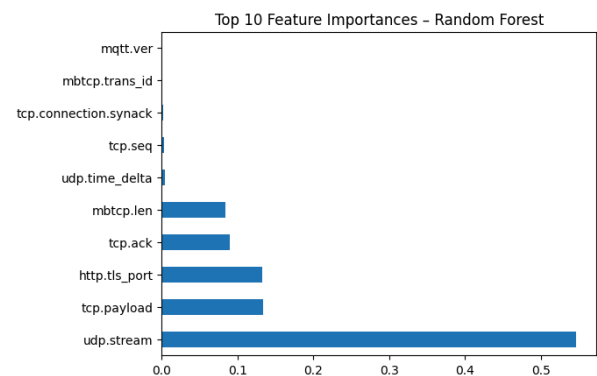


Fig.5. Feature Importance - Random Forest

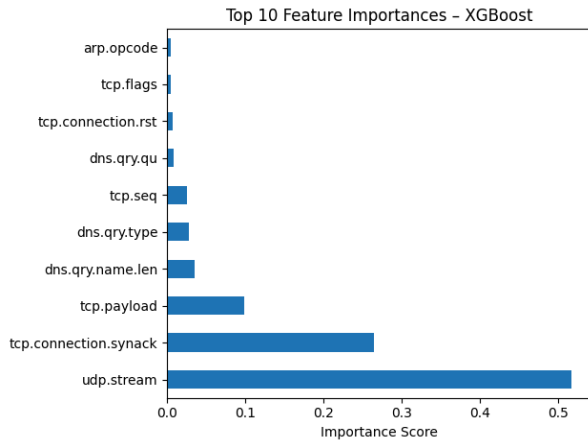


Fig.6. Feature Importance - XGBoost

V. CONCLUSIONS

A supervised learning-based intrusion detection system for real-time detection of UDP Flood and ARP Spoofing attacks in IoT and IIoT networks was discussed in the proposed research work. A comprehensive, multi-dataset experimental approach was employed to design and test five supervised machine learning classifiers: Logistic Regression, Support Vector Machine, Random Forest, Gradient Boosted Decision Trees i.e. XGBoost and a simple Multi-Layer Perceptron. The efficiency of supervised learning algorithms in a controlled environment was also verified by experimental evaluation on the Edge-IIoTSet dataset, which demonstrated near-perfect detection performance for all models. Nevertheless, this work provides a critical and realistic explanation of these results rather than accepting them as evidence of perfect detection power. The high separability of features and bias in benchmark intrusion detection datasets, where flow-level features directly represent attack activity, are the primary reasons for the observed accuracy. Supervised learning algorithms are capable of learning highly discriminative decision boundaries based on these features, which lead to near-perfect performance.

Cross-dataset validation was employed to further evaluate the proposed models in order to validate this observation and avoid the risk of making wrong conclusions. The evaluation of the BETH honeypot dataset showed a significant drop in performance without retraining, with many anomalies triggered. This demonstrated how detrimental a domain shift can be and the limited ability for generalization. The results of these experiments show that an evaluation of robustness is necessary and that high accuracy on the benchmark is not enough evidence to prove practical intrusion detection performance. The proposed work highlights an important area for current IDS research, the gap between achieving success in an operational environment and achieving success on a benchmark by combining performance evaluation across datasets with strong in-domain evaluation. It is essential to measure IDS evaluations based on transparency and secondary validation as well as consideration of domain awareness rather than solely focusing on accuracy as the primary goal.

In conclusion, while supervised learning models are extremely effective at detecting ARP Spoofing and flooding attacks against UDP within a laboratory setting, deploying a model requires a detailed analysis of an organization's physical environment and the possibility of bias in the data used to develop the model. Future research will focus on how to overcome these challenges and create more robust intrusion detection systems in real-world IoT environments through domain adaptation, innovative combinations of anomaly-classification models, ongoing learning, and updating of models in real time.

VI. REFERENCES

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